B.TECH · LABORATORY DEMONSTRATION

Fusion Splicing, OTDR Testing & Link Loss Measurement

Optical Fiber Communication Laboratory

A Fusion splicing **B** OTDR testing and loss measurement

TEAM MEMBERS

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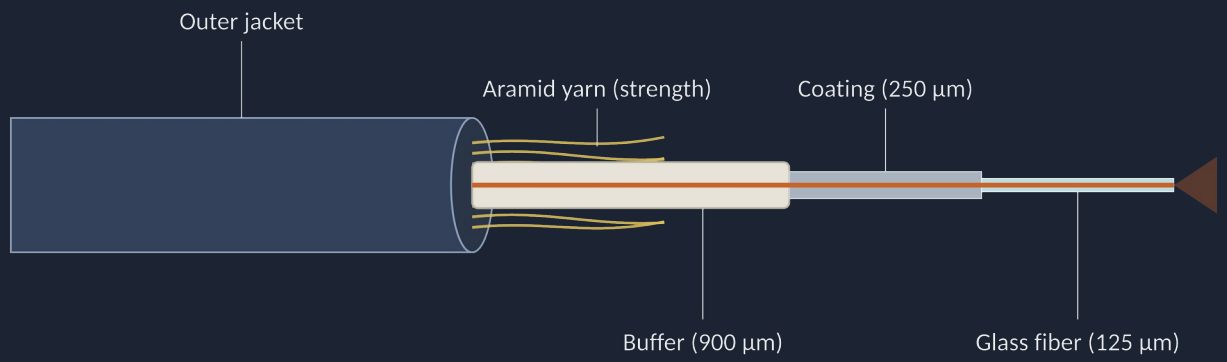
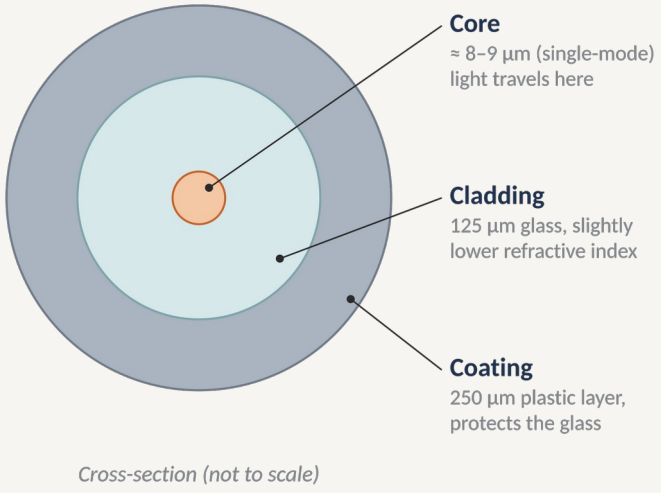
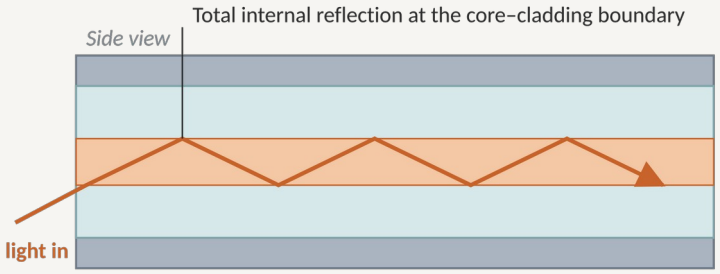


Illustration — layers not to scale

Optical Fiber and Fusion Splicing



Cross-section (not to scale)



Core index n_1 is a little higher than cladding index n_2. Light hitting the boundary at a shallow angle bounces back into the core, so it stays trapped and travels long distances.

Light travels in the tiny core. The cladding has a slightly lower refractive index, so light keeps reflecting back into the core (total internal reflection).

What is splicing?

Joining two fibers end-to-end so light passes from one core into the other.

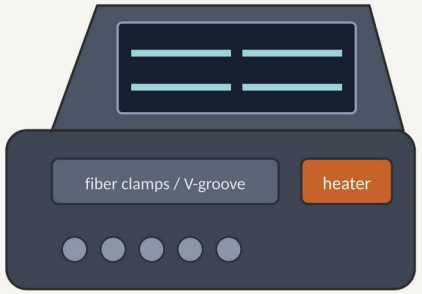
Fusion splicing

An electric arc melts the two glass ends into one piece. The joint is permanent, with very low loss (typically below 0.1 dB).

Why alignment matters

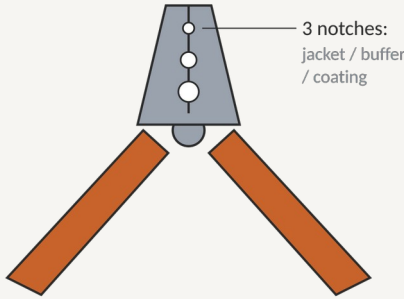
The core is only about 9 µm wide. Dirty, angled or offset ends lose light at the joint.

Equipment Required for Fusion Splicing



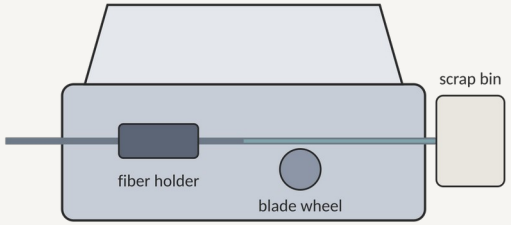
Fusion splicer

Aligns the two fibers with cameras and motors, then fuses them with an electric arc.



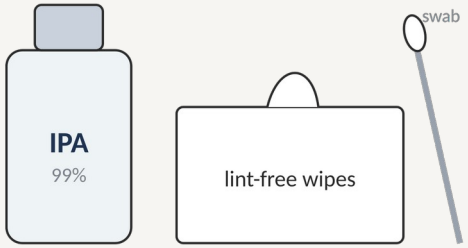
Fiber stripper

Removes the jacket, buffer and 250 µm coating without scratching the glass.



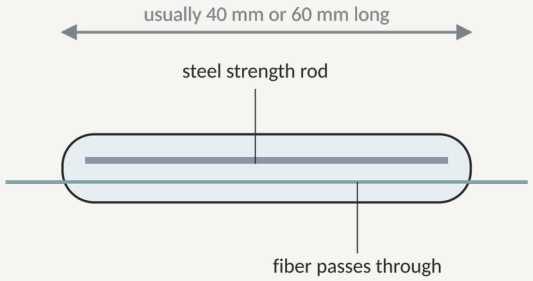
Precision cleaver

Cuts the bare fiber so the end is flat and at 90° to its length.



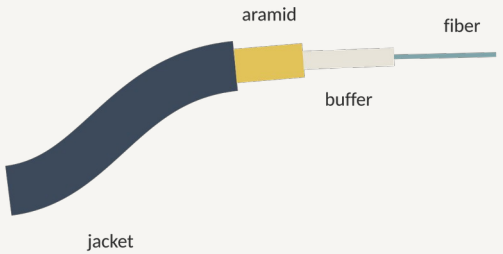
Cleaning kit

99% isopropyl alcohol and lint-free wipes remove dust and oil from the glass.



Heat-shrink sleeve

A tube with a steel rod that shrinks over the joint to protect it.



Optical fiber cable

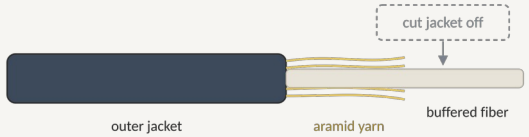
The fiber to be joined; check its type before you start.

Fusion Splicing Procedure (1/2)



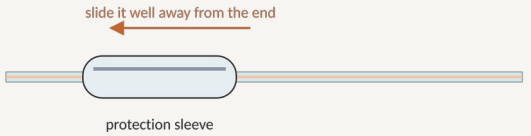
1 Prepare the cable

Check the fiber type and mark how much jacket to remove.



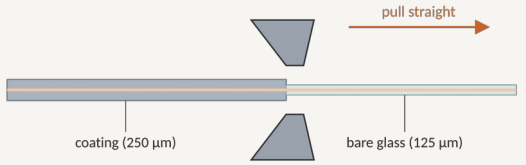
2 Remove the jacket

Strip the outer jacket; cut the aramid yarn with scissors.



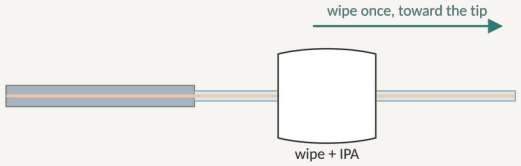
3 Slide on the sleeve

Put the heat-shrink sleeve on one fiber now — it can't be added later.



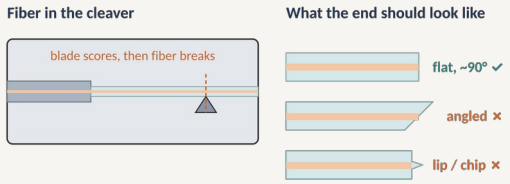
4 Strip the coating

Remove the 250 µm coating in one smooth pull (about 3–4 cm).



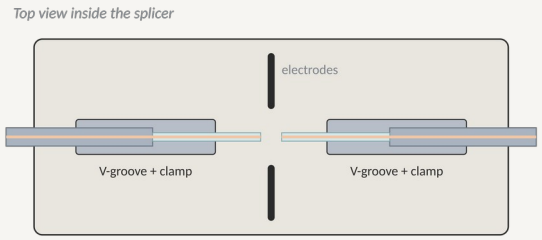
5 Clean the fiber

Wipe the bare glass once with 99% IPA on a lint-free wipe.



6 Cleave the ends

Cut a flat, square end (within about 1°). Bin the offcut.

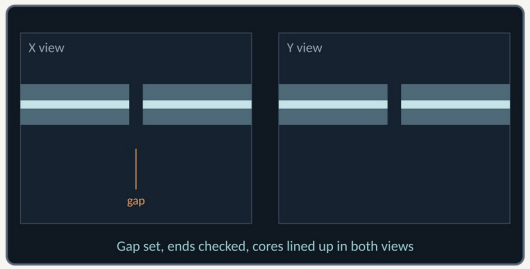


7 Load the splicer

Lay both fibers in the V-grooves, tips near the electrodes. Close the clamps.

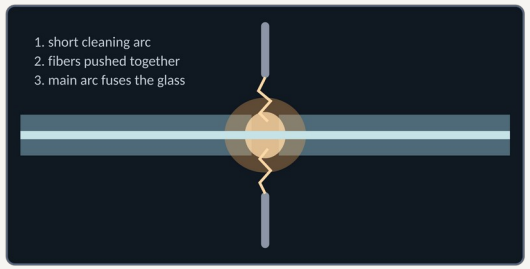
Safety: bare-fiber offcuts are tiny glass needles. Pick them up with tape and put them in the scrap bin.

Fusion Splicing Procedure (2/2)



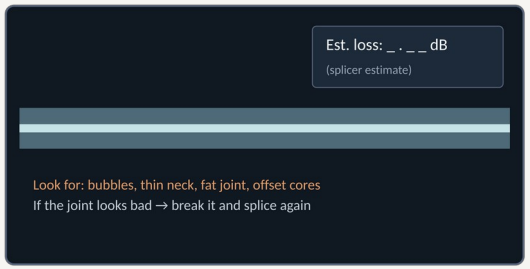
8 Align the cores

Cameras view the fibers in X and Y; the splicer lines up the cores.



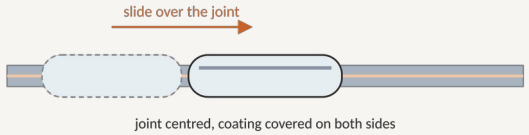
9 Fuse with the arc

A short cleaning arc, then the main arc melts the ends into one.



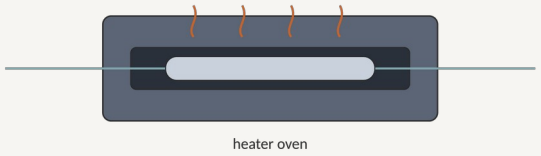
10 Inspect the splice

Check the joint on screen. The loss shown is only an estimate.



11 Move the sleeve

Slide the sleeve so the joint sits in its centre.



12 Heat-shrink

Shrink the sleeve in the heater. Let it cool before handling.



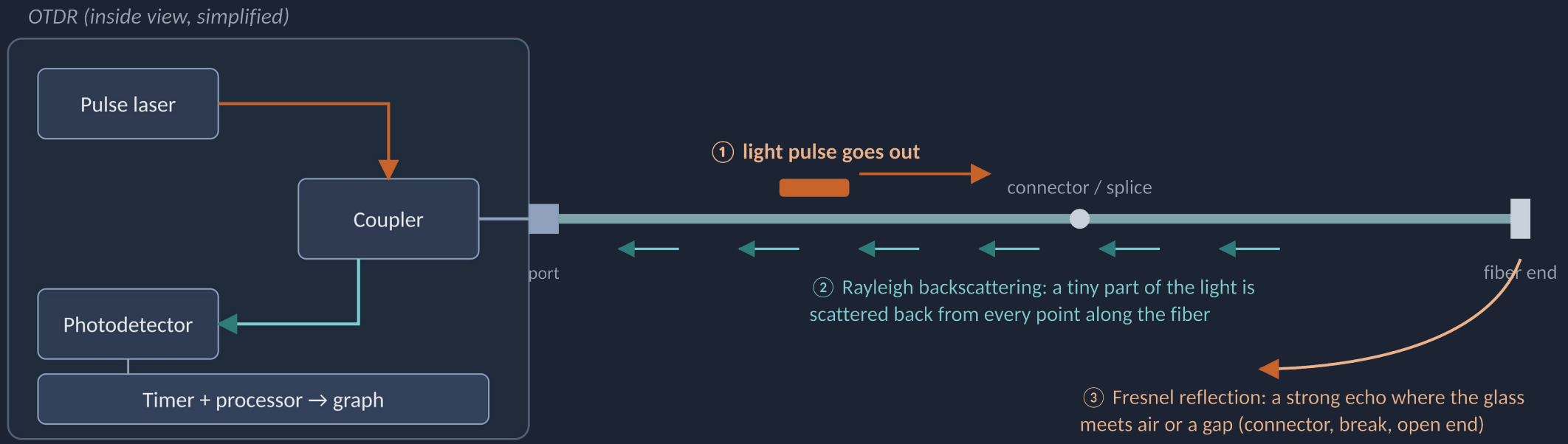
13 Test the joint

Measure the real loss with an OTDR or a power meter.

The splicer's loss value is an estimate from its camera image. Record the measured loss from step 13. (Screen views are illustrations; we did not record a splice-loss value.)

What Is an OTDR and How It Works

OTDR = Optical Time Domain Reflectometer. It finds the **length, losses and faults** of a fiber from just one end — it sends light pulses and times the light that comes back.

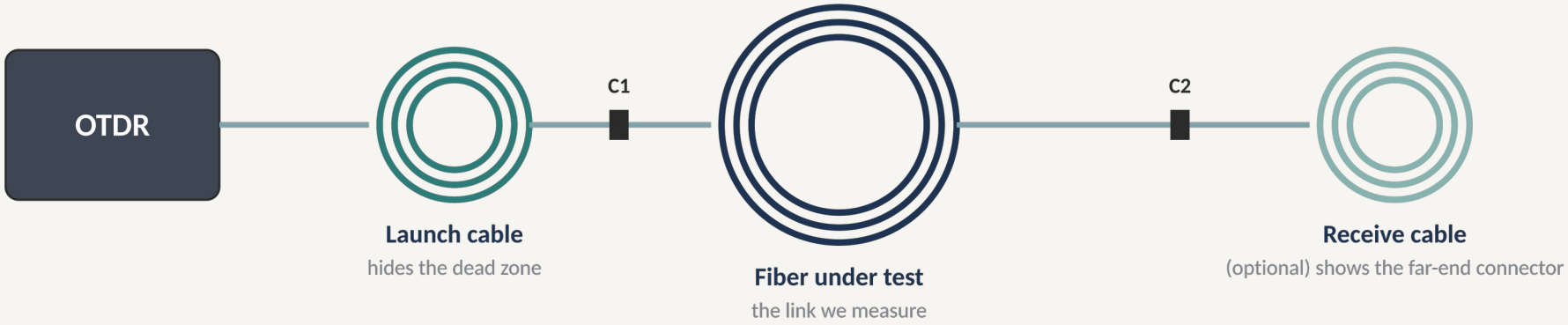


④ **Time of the echo → distance:** $d = (c \times t) / (2\,n)$

c = speed of light in vacuum, t = round-trip time, n = group index of the fiber (≈ 1.47). Divide by 2 because light goes out and back.

Connecting and Running the OTDR Test

Connection order



Connect

1. Switch on the OTDR.
2. Inspect and clean all connectors.
3. Connect launch cable (if available), then the fiber under test.
4. Check connections are firm.
5. Make sure no live signal is on the fiber.

Run the test

1. Open the Test Setting menu.
2. Set wavelength, range, pulse width, time, resolution, power mode.
3. Return to the main screen.
4. Press AVG to start the averaged test.
5. Wait, then read the trace and event table.

Why launch / receive cables?

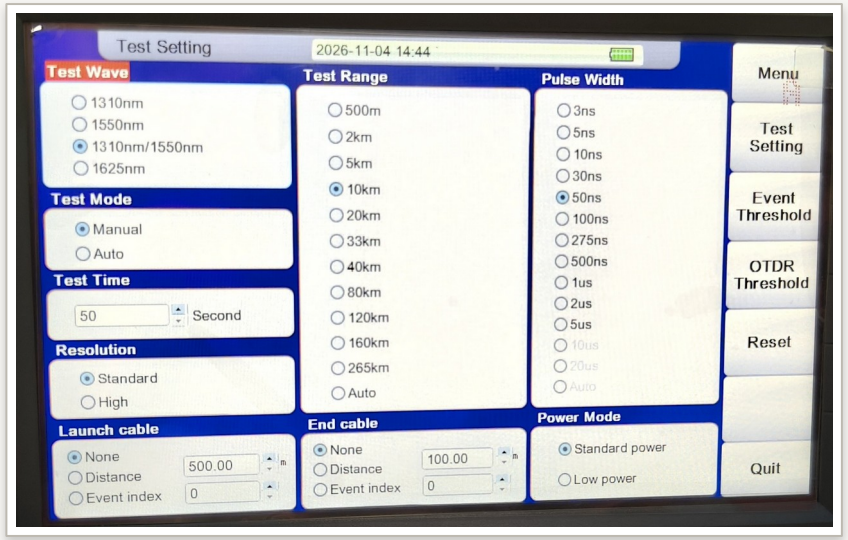
The strong reflection at the OTDR port hides the first few metres (dead zone). A launch cable moves the first connector out of it; a receive cable lets the last connector be measured.

⚠️ **Never look into the OTDR port or a fiber end — the laser light is invisible.**

OTDR Test Settings We Used

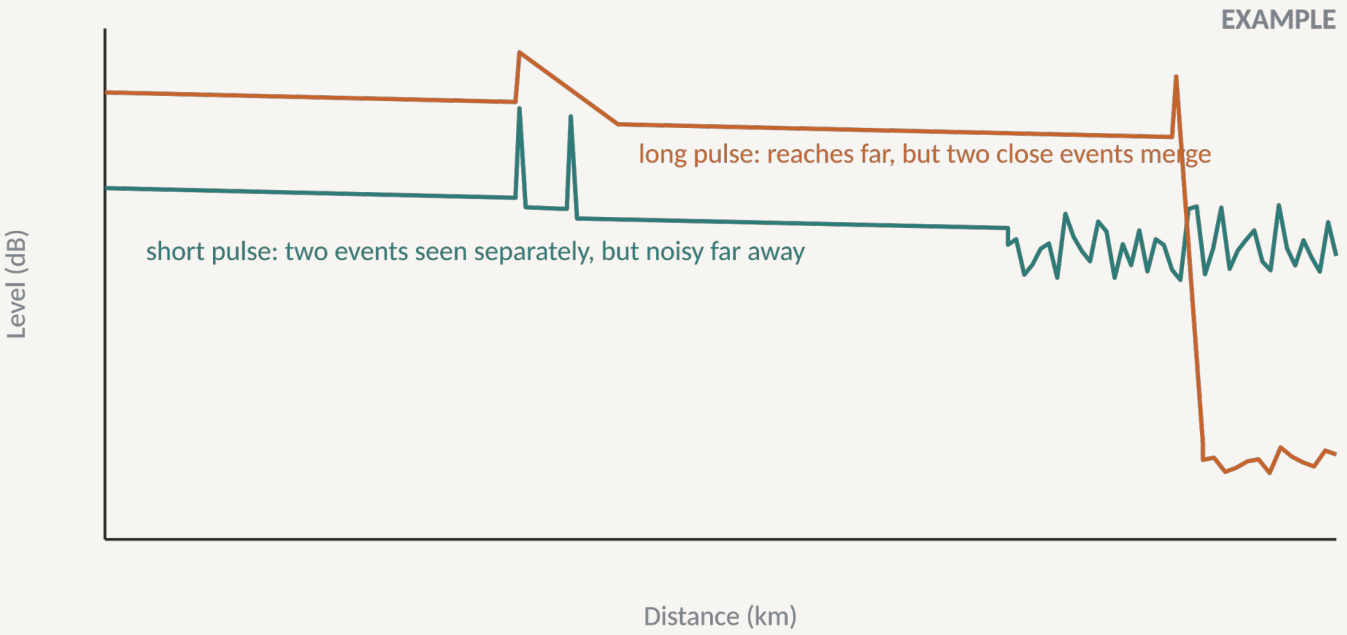
Parameter	Our value	What it means
Test wavelength selection	**1310/1550 nm**	Wavelength(s) of laser light used for the test
Test range	**10 km**	Longest distance shown; set longer than the fiber
Pulse width	**50 ns**	Pulse length: short = fine detail, long = reaches farther
Test mode	**Manual**	We chose the settings ourselves (not Auto)
Test time	**50 seconds**	Averaging time: longer gives a smoother trace
Resolution	**Standard**	How finely the trace points are spaced
Launch cable setting	**None**	No launch cable length entered or removed
End cable setting	**None**	No receive (end) cable entered
Power mode	**Standard**	Instrument power option, left at default

Short pulse + long averaging = a clear, detailed trace for a short (few-km) fiber like ours.



Values copied exactly from our settings screen (Photo 1).

Pulse Width, Resolution and Averaging



Our settings: 50 ns pulse, 10 km range, 50 s averaging → about 5 m resolution and a clean trace for our 4.8 km fiber.

Resolution ≈ c·τ / 2n (τ = pulse width)

50 ns: $3\times10^{8} \times 50\times10^{-9} / (2 \times 1.468) \approx 5.1$ m

Short pulse

Fine detail: close events are seen separately, short dead zones. But less light goes in, so the far end is noisy.

Long pulse

More energy, so it reaches much farther (more dynamic range), but close events merge into one.

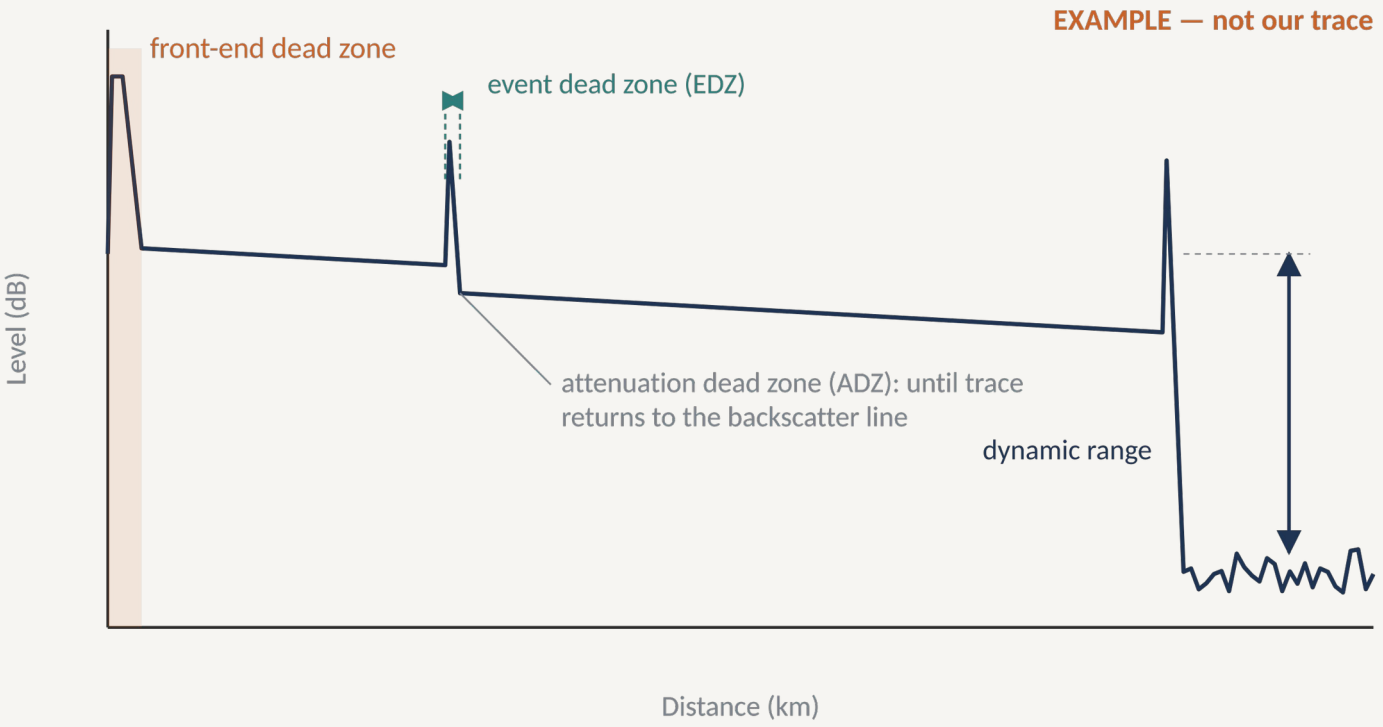
Averaging (test time)

Noise falls with √N: four times more averages roughly halves the noise. Longer test time = smoother trace.

Range

Must be longer than the fiber, or echoes from one pulse overlap the next and false events appear.

Dead Zones and Dynamic Range



In our test the launch cable was set to 'None', so the OTDR's own front connector sat inside the front-end dead zone and its loss could not be measured.

Event dead zone (EDZ)

Shortest distance after a reflection at which a second reflective event can be seen as separate.

Attenuation dead zone (ADZ)

Distance after a reflection before the loss of a next event can be measured. Always longer than the EDZ.

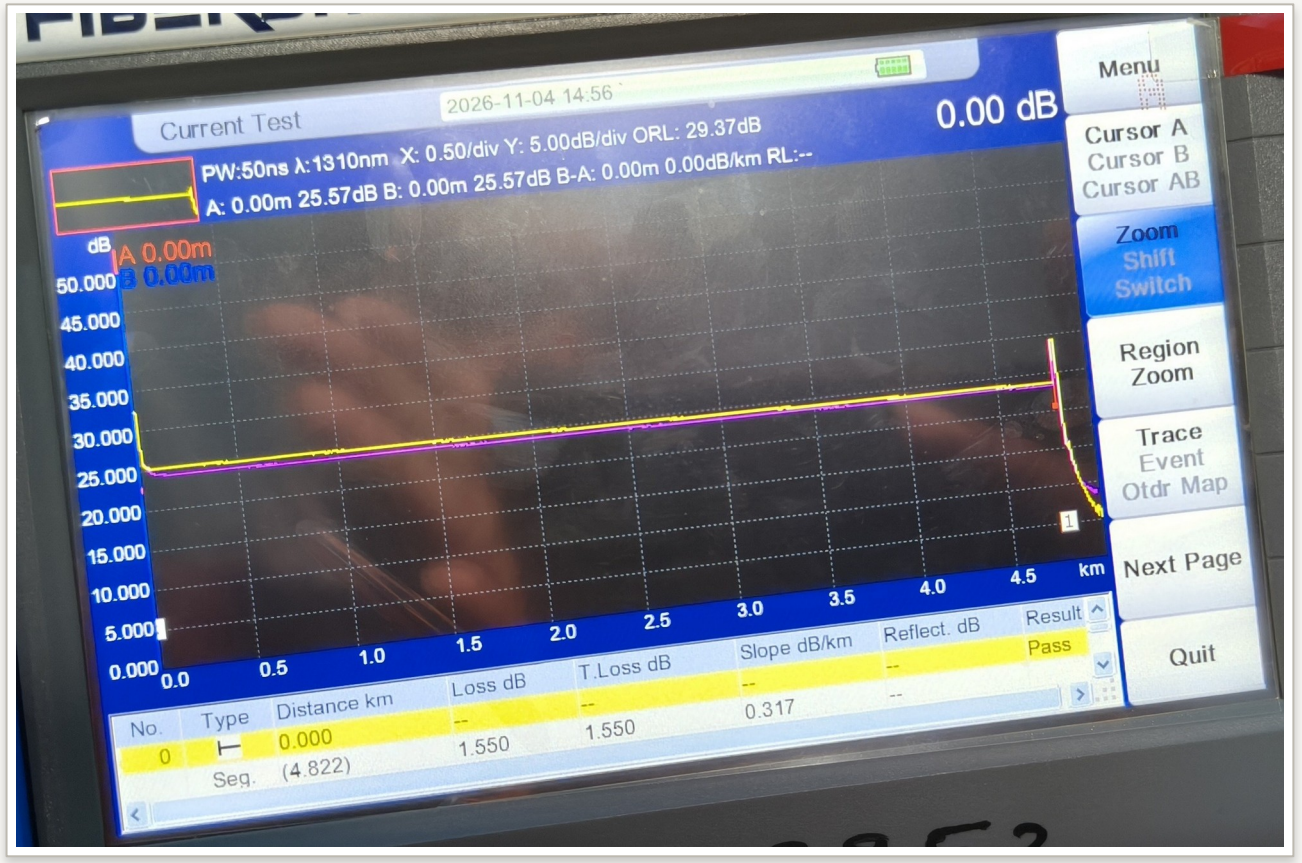
Dynamic range

dB gap between the backscatter level at the start and the noise floor. It decides the largest link loss the OTDR can measure — keep a few dB of margin.

How to beat dead zones

Use a launch cable, shorter pulses and clean connectors.

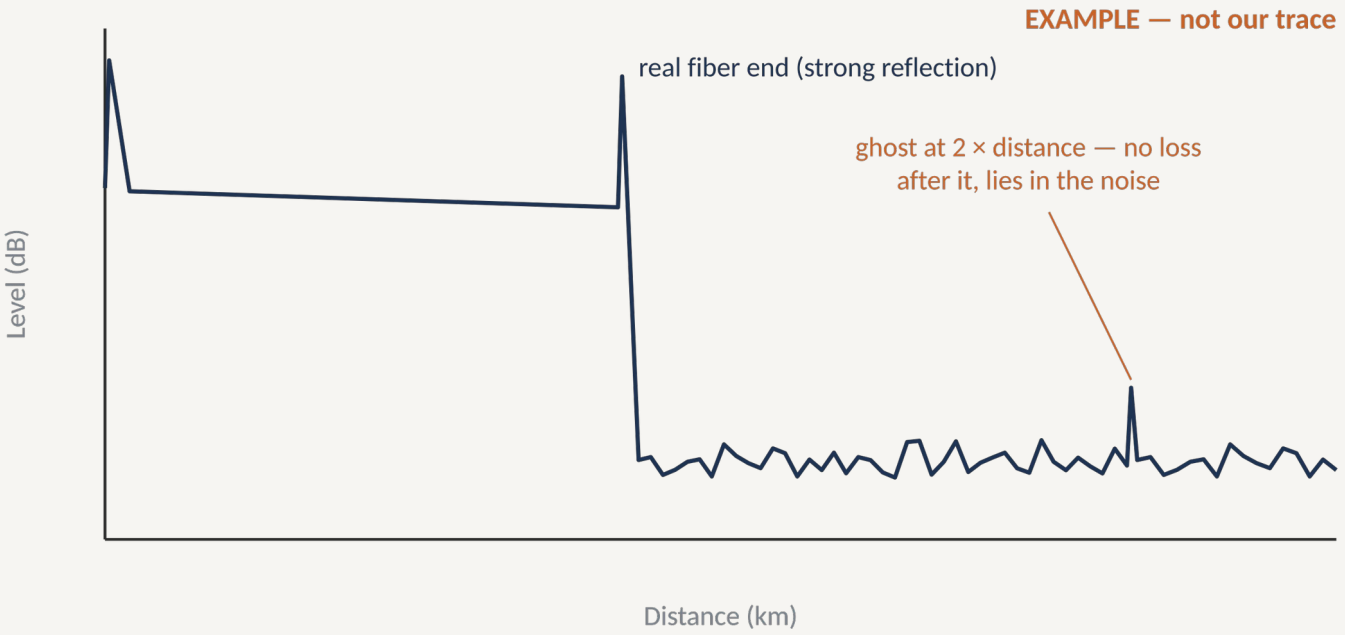
Understanding the OTDR Graph



- **X-axis**
  Distance along the fiber, in kilometres.
- **Y-axis**
  Level of the returned signal, in dB.
- **Sloping straight line**
  Normal fiber attenuation — the light weakens a little every km.
- **Sudden downward step**
  A loss event: possibly a splice, bend or other loss point.
- **Upward spike**
  A reflection: usually a connector or the fiber end.
- **Sharp drop to noise**
  End of the measured fiber link.

We only mark events that the OTDR actually shows on our screen and event table. We do not guess faults that are not on the trace.

Reflectance, ORL and Ghost Events



In our trace

- The event table shows one fiber section: start at 0.000 km, length 4.822 km, loss 1.550 dB — no splice or bend was flagged in between.
- A large spike at about 4.8 km is the reflection from the far end of the fiber.

Reflectance (one event)

How much light a single connector or end reflects, in negative dB. Typical: open cleaved end ≈ −14 dB, good UPC connector −50 dB or lower, APC −60 dB or lower.

ORL (whole link)

Launched power ÷ total reflected power, in dB. Higher is better. Ours: 29.37 dB, most likely set by the strong far-end reflection.

Ghost events

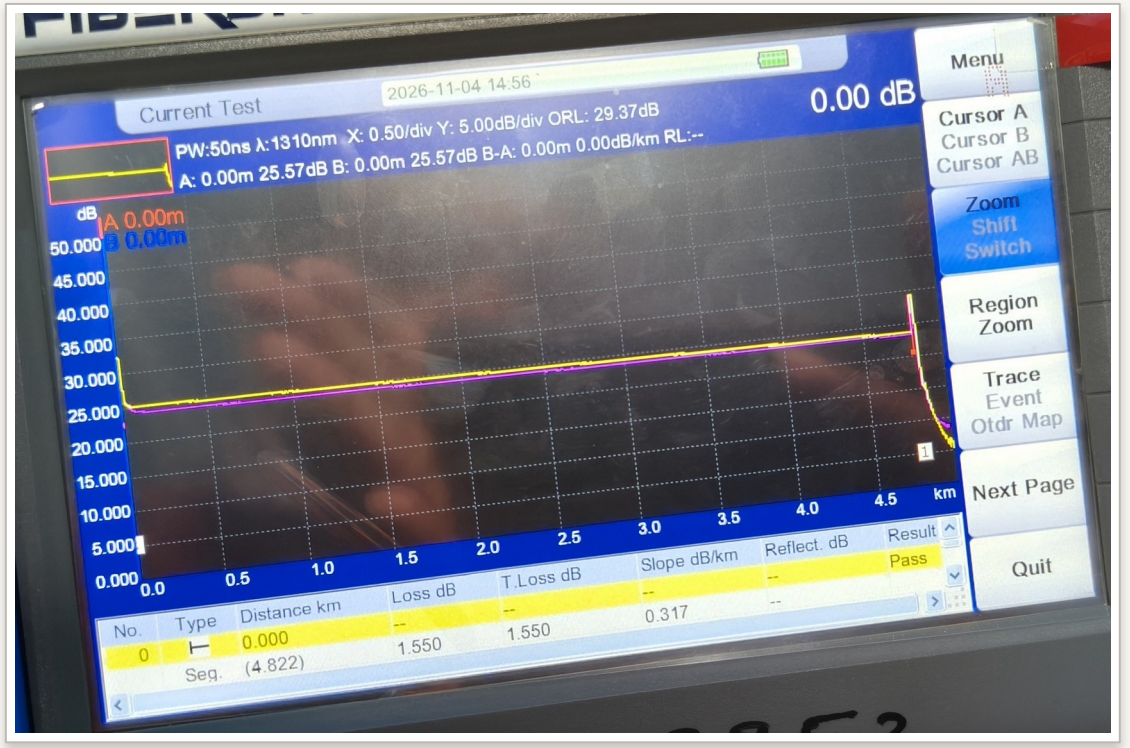
False echoes of a strong reflection, at 2× its distance. No loss across them and they often sit in the noise. Fix: clean or terminate connectors, change the range.

Our OTDR Observations

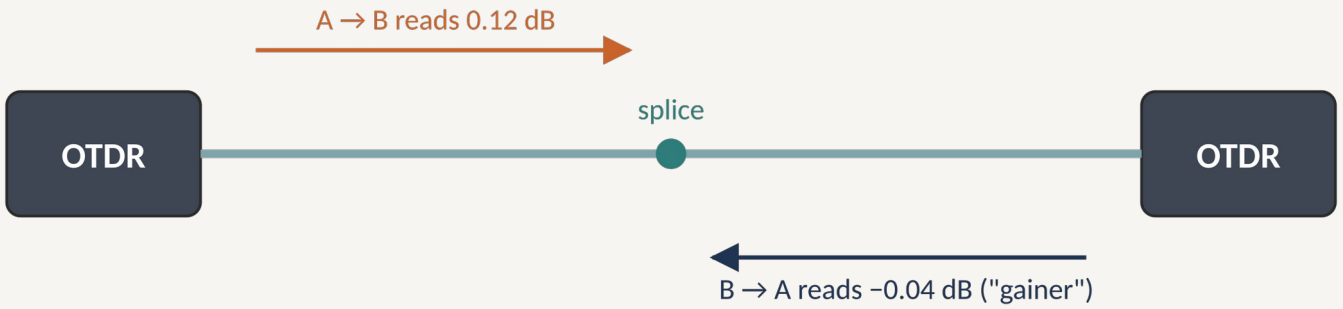
Parameter	Observed value
Test wavelength	**1310 nm**
Pulse width	**50 ns**
Fiber distance	**4.822 km**
Total loss	**1.550 dB**
Displayed attenuation slope	**0.317 dB/km**
Optical Return Loss (ORL)	**29.37 dB**
Instrument result	**PASS**

All values read directly from the result screen (Photo 2). Nothing here is calculated.

What PASS means: the measured results were within the thresholds set in the instrument. It is not proof that the link meets every possible performance requirement.



Two Wavelengths and Bidirectional Testing



true loss ≈ (0.12 + (−0.04)) / 2 = 0.04 dB

Example numbers for illustration only

Why one direction is not enough

Two joined fibers can scatter back slightly different amounts of light (for example, different mode-field diameters). Seen from one side a splice can look worse than it is, or even show a gain (a 'gainer'). Testing from both ends and averaging cancels this. It is the standard way to report splice loss.

1310 nm vs 1550 nm

1550 nm has lower fiber loss (typically ≈ 0.2 dB/km vs ≈ 0.33 dB/km at 1310 nm), but it is much more sensitive to bends.

Finding bends

If an event's loss is much larger at 1550 nm than at 1310 nm, it is likely a bend. The same at both is likely a splice or connector.

1625 nm

Also on our OTDR menu. Used to test fibers in service (with filters), without disturbing traffic.

Our next step

Our result screen is at 1310 nm; repeating at 1550 nm and from the other end would complete the test.

Results and Conclusion

4.822 km

fiber length · at 1310 nm

1.550 dB

total loss · at 1310 nm

PASS

instrument result · at 1310 nm

Discussion

- Average attenuation ≈ 0.321 dB/km (calculated); OTDR slope 0.317 dB/km.
- PASS only means the results were inside the instrument's set thresholds.
- Launch/end cables were 'None', so the first and last connector losses are not accurate.

Conclusion

1. Fusion splicing joins fibers permanently using an electric arc; clean, square cleaves give low loss.
2. An OTDR measures length and loss and locates faults from one end of the fiber.
3. Link loss is read from the OTDR trace; our link: 1.550 dB over 4.822 km at 1310 nm.

References: G. Keiser, Optical Fiber Communications (McGraw-Hill) · J. M. Senior, Optical Fiber Communications (Pearson) · FOA Online Reference Guide, thefoa.org/tech/ref · ITU-T G.652 · TIA-568.3-D · lab OTDR and splicer manuals.

Thank You

Questions?

Presented by **Neelesh** · **Sriyans** · **Abhilash**

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